

Work Term Report:

Exploring the Role of Data Analysts in Digital Scholarship -

Preserving Academic Knowledge in the Digital Age

Shirling Liu (4th Year Statistics Co-op)

Work Term #1: Fall 2023 – Winter 2024 (8 Months)

UTSC Library Digital Scholarship Unit

Data Analyst Co-op

Table of Contents

Executive Summary.....	3
Introduction.....	4
An Overview of the Work Term	5
Analytical Section	7
<i>Work Term placement</i>	<i>7</i>
<i>Work Term Duties and Objectives</i>	<i>7</i>
<i>Comparing Data Visualization Tools: Power BI vs. Looker Studio</i>	<i>9</i>
<i>Interpersonal and Technical Skills Required for a Successful Work Term</i>	<i>10</i>
<i>Interpersonal and Technical Skills Improved throughout this Work Term</i>	<i>11</i>
<i>Application of Work Term to Future Career.....</i>	<i>12</i>
Conclusions and Recommendations.....	14
Appendices.....	15
Bibliography	16

Executive Summary

During my first work term, I held the position of a data analyst at UTSC Library's Digital Scholarship Unit (DSU) from September 2023 to April 2024 (8 months). As a part of the unit at UTSC Library, DSU supports the digital scholarship initiatives of UTSC, which include data management, data preservation, digital humanities, data visualization, and scholarly communication. My primary responsibility was utilizing data-driven insights to enhance these initiatives. This experience has significantly enriched my understanding of DSU's mission. It has allowed me to apply theoretical knowledge from the university in real-world applications, contributing significantly to my professional development.

Introduction

The purpose of writing this work term report is two-fold: first, to provide a comprehensive overview of my experiences and achievements while being a data analyst at the Digital Scholarship Unit, and second, to document what I have learned during this eight-month internship and the ability to gain refinement. The rationale for these goals is rooted in the need to not only document my contributions and growth but also to critically analyze the impact of my work on organizational goals. This report aims to comprehensively understand my role and its importance within the broader context of the UTSC Library's operations by achieving these objectives. Through reflection and analysis, it attempts to elucidate the rationale behind my critical decisions and actions throughout my career, ultimately contributing to a deeper understanding of the challenges, successes and opportunities encountered.

An Overview of the Work Term

While working in the UTSC Library's Digital Scholarship Unit (DSU), I served as a data analyst, contributing to the department's mission of advancing digital research and learning initiatives. As mentioned on DSU's website: 'The University of Toronto Scarborough Library's Digital Scholarship Unit (DSU) provides UTSC-specific support for digital scholarship in the areas of data, the digital humanities, and digital collections building, digital preservation, and scholarly communications'¹. DSU is a central hub for innovative programs designed to leverage digital tools and resources to support academic endeavours. My role at DSU covers various responsibilities, from data collection, analysis, and visualization to developing data-driven solutions to support academic research.

As a data analyst, my main responsibility is to collect, clean and analyze data from different sources, including databases (such as Google Analytics and SharePoint List), spreadsheets (such as Excel and CSV files) and external application programming interfaces (APIs). To process large data sets and use state-of-the-art statistical techniques to derive meaningful information, I use data visualization tools such as Power BI and Google Looker Studio to explain trends, patterns, or correlations in charts.

In addition to refining and implementing analytical models and algorithms to optimize data trends, I collaborate with a multidisciplinary team within DSU, including website developers and engineers, to translate requirements for optimizing website traffic into

actionable data-driven solutions. Furthermore, I actively engage in meetings and presentations with professors and work-study students from the Departments of Biological Sciences and Departments of Arts, Culture, and Media, effectively communicating findings and recommendations. Recognizing the need for users to become more familiar with data cleaning and visualization, I prioritize assisting them in cleaning databases and creating interactive dashboards. These dashboards offer engaging platforms for exploring academic content. Simultaneously, I actively engage in external training courses and utilize online resources to bolster my technical proficiency in data manipulation, programming languages such as HTML and SQL, and Data Visualization Tools. These experiences not only refine my data analysis skills but also reinforce my dedication to leveraging data-driven insights to propel academic success at DSU.

Overall, my work term at the UTSC Library's Digital Scholarship Unit provided me with invaluable hands-on experience in data analysis, allowing me to apply theoretical knowledge gained from coursework to real-world scenarios. It deepened my understanding of digital scholarship and fortified my commitment to utilizing data-driven insights to support academic research and learning initiatives within the university community.

Analytical Section

Work Term placement

I spent my first work term at the UTSC Library Digital Scholarship Unit for 8 months. Being a member at UTSC, starting my career in a familiar environment allowed me to adapt to the workspace more quickly. Working for the university where I studied has provided me with a deep understanding of its culture and policies and a network of familiar faces, facilitating a seamless transition into my role and offering opportunities for professional growth. Moreover, the internship opportunity provided by UTSC aimed to teach us how to integrate into the workspace environment while focusing on improving the necessary skills. Therefore, my supervisor and colleagues showed great patience and kindness, always ready to teach me and inquire about unfamiliar tasks while also seeking feedback to help me improve my skills and adapt more effectively. They assigned tasks based on my feedback, and the weekly meetings helped me understand my colleagues' work better, facilitating better coordination. I believe this experience is an ideal opportunity to integrate what I have learned in university with real-world practicality in an educational-professional setting. This experience not only deepened my understanding of digital scholarship but also allowed me to learn more about the university I attended for four years by becoming part of its operations.

Work Term Duties and Objectives

As a data analyst within DSU, my central focus is enhancing librarians' grasp of

collected data and users' understanding of academic knowledge. This encompasses daily responsibilities such as ensuring data quality and conducting necessary cleaning procedures. Additionally, I generate a variety of reports using Power BI, notably the UTSC Library Website Traffic reports (Figure 1). This involves collecting, cleaning, and analyzing website traffic data to provide insights for developers and engineers, ultimately optimizing search engine performance.

I work with various data sets from colleagues, preserve important digital information for library and UTSC members, and visualize this data to help colleagues improve the quality and efficiency of their work. My overarching objective is to craft user-friendly reports that not only enhance colleagues' effectiveness but also transform digital data into concise, comprehensive summaries.

Moreover, I thrive in collaborative environments, working extensively with professors and work-study students across various departments. Together, we develop dashboards for K-12 and other students, facilitating accessible and engaging research exploration. For instance, I collaborated with professors and work-study students from the Department of Biological Sciences to create an interactive dashboard that teaches K-12 students, and anyone interested in biology about biology concepts portrayed in movies. This involved searching for movies correlated with multiple disciplines, determining scientific portrayals, claims, and verdicts, and transforming data for visualization using Looker Studio (Figure 2). This collaborative

effort exemplifies DSU's mission to leverage digital tools and resources to support academic endeavours, and it has honed my technical and interpersonal skills, which are crucial for my future career aspirations.

Comparing Data Visualization Tools: Power BI vs. Looker Studio

Throughout this work term, I have experience using Power BI and Looker Studio. I mostly use Power BI as a visualization tool for the UTSC Library, while Looker Studio is utilized for projects with professors. The reason why I have these two tools in different ways is due to their characteristics. Power BI, a data visualization tool developed by Microsoft, allowed users to create reports by connecting to multiple disciplines of data sources. It has an online workspace for organizations to organize their data and connect the various sources by creating data. In this case, only members inside the organization (UTSC Library) can view the report, which means that it posits security and confidentiality. However, it has the limitation to be a proper tool to spread knowledge to the public.

On the other hand, Looker Studio is an online visualization tool developed by Google, with the limitations of only connecting the data source with Google systems and the lack of confidentiality. However, it can create an interactive report to spread the research to the public. In this case, it is essential to determine the data's source and the report's purpose. Although they are both software for data visualization, they all have their pros and cons, and there is no perfect software; the most important thing is finding the most suitable software.

Interpersonal and Technical Skills Required for a Successful Work Term

A dynamic set of skills and tools is required to effectively fulfill the job duties of a data analyst in the Digital Scholarship Unit. Firstly, analytical and research skills are essential. Since the primary responsibility of a data analyst is to analyze data from various sources, it is necessary to deal with different types of data and utilize analytical skills to gain insights into valuable data and make professional decisions regarding optimized data performance and visualization. Strong research skills are crucial for finding new approaches and possible solutions when encountering difficulties or obstacles.

In our remote work format, self-learning skills are not just necessary but empowering. When faced with new tasks, self-learning is often necessary, and remote working environments can make contacting colleagues for urgent tasks challenging. However, the ability to independently research online and seek guidance from supervisors or colleagues when available makes me capable of handling any situation. Additionally, the capacity to quickly learn and adapt to new systems is essential and a testament to my adaptability and efficiency in work execution.

Finally, excellent verbal and written communication skills are not just crucial; they are the key to our effective collaboration. Effective communication is more challenging in virtual environments, but solid verbal communication skills ensure that colleagues and supervisors understand my explanations and concepts. Written communication is also essential for tasks

that require a deeper understanding of data sources, report purposes, and colleague requirements, often conducted through Teams chat or email. Clear and concise communication enhances our work efficiency and confidence in conveying my ideas.

As co-op students, we should leverage our positive skills to make meaningful contributions to the organization while absorbing as much experience and knowledge as possible from seasoned professionals during the work term. I have effectively applied my refined skills to support DSU and have consistently performed to the best of my ability. Additionally, I have gleaned valuable insights and knowledge from my colleagues, resulting in a mutually beneficial arrangement for both parties.

Interpersonal and Technical Skills Improved throughout this Work Term

In 8 months of working, I have seen considerable growth in my interpersonal and technical skills. First and foremost, my communication skills have notably improved. The workplace demands constant interaction with colleagues, making it an ideal environment to practice and refine verbal and written communication skills.

Teamwork has also become a crucial aspect of my work. Collaborating across different departments and with colleagues has strengthened connections and significantly enhanced the quality and efficiency of our completed tasks. Initially, I took a long time to become familiar with a new project as I needed to adjust to different working styles and thought processes. However, through participation in various projects alongside coworkers, I have developed the

ability to coordinate work effectively within a team environment.

Furthermore, my proficiency in data analysis has seen significant advancement. Working with multiple databases has allowed me to explore various data collection and analysis methods. For instance, one project I worked on involved collecting webforms submitted by students, staff, professors, and external individuals on the UTSC Library website. Initially, data collection has been relied on manually checking separate submissions on each website, leading to inefficiencies. Collaborating with my supervisor, we devised solutions to streamline the process and create a comprehensive report. This experience allowed me to research different data collection methods and analyze HTML information, improving my analytic skills. In summary, these improvements in communication, teamwork, and data analysis skills have strengthened my abilities and contributed to the DSU.

Application of Work Term to Future Career

As mentioned earlier, I have developed my communication, teamwork, and analytical skills during this work term, which I consider valuable for my future career. However, one aspect I have not highlighted but find equally important is the growth of my confidence, both in the workplace and in my daily life. While my supervisor noted room for improvement in this area during the final evaluation—acknowledging that it is a common challenge for those starting their first job—I believe I have made strides.

I have gradually gained confidence throughout the work term, understanding that it is a

continuous process. While there is still ample room for improvement, I am optimistic about becoming more confident, particularly as I gain more experience in subsequent work terms. Additionally, my role as a data analyst has provided insight into potential career paths post-graduation. This work term has illuminated a promising trajectory for my future career. It has allowed me to navigate the workplace with increased confidence and seamlessly integrate into its dynamics.

Conclusions and Recommendations

In conclusion, my work term at the UTSC Library Digital Scholarship Unit has been pivotal in my personal and professional development. Starting my career within the university provided a familiar environment, facilitating a seamless transition and offering opportunities for growth. The internship emphasized the importance of integrating into the workplace while honing essential skills, supported by patient guidance from supervisors and colleagues. Weekly meetings and tailored feedback enhanced my understanding of team dynamics and project objectives, fostering better coordination and productivity.

Responsibilities as a data analyst within DSU offered diverse challenges, from ensuring data quality to producing insightful reports. Collaborative efforts with professors and work-study students exemplified DSU's mission to support academic endeavours through digital tools, refining technical and interpersonal skills.

Looking forward, I believe the skills and experiences gained during this work term will shape my future career trajectory. Beyond technical proficiency, confidence and adaptability will be essential assets in navigating the workplace. As I continue to build upon these foundations, I am optimistic about the opportunities ahead and confident in my ability to make meaningful.

Appendices

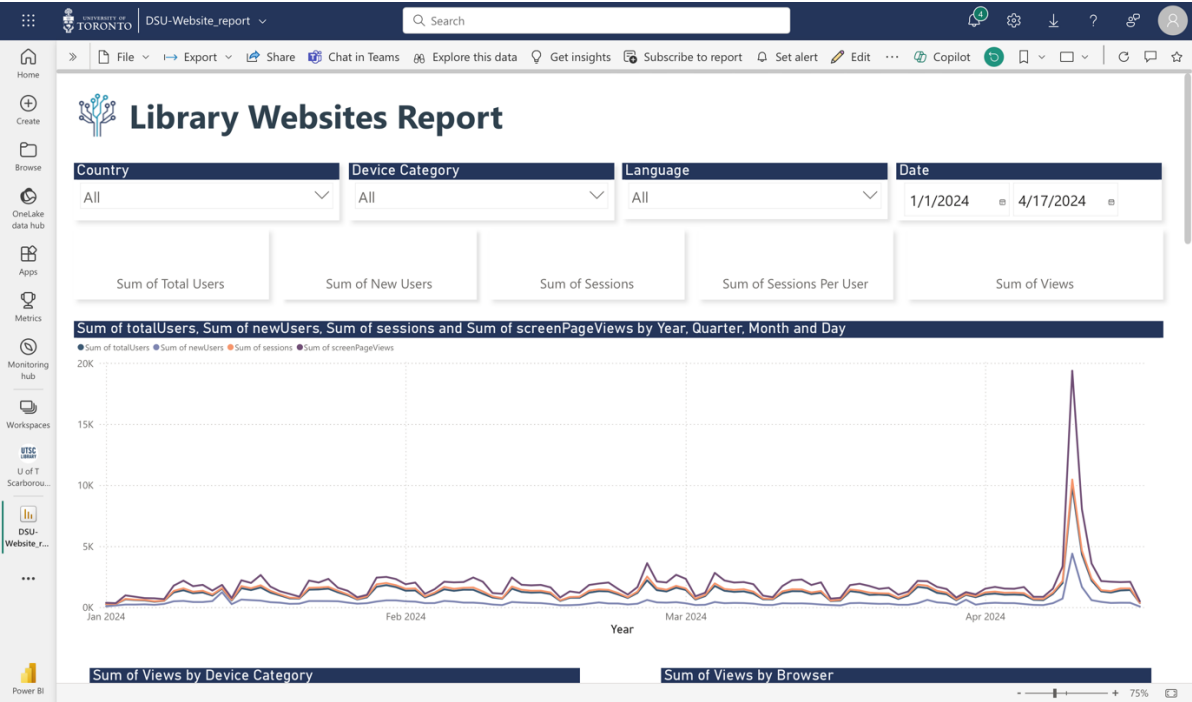


Figure 1. UTSC Library Website Traffic Report (Power BI).

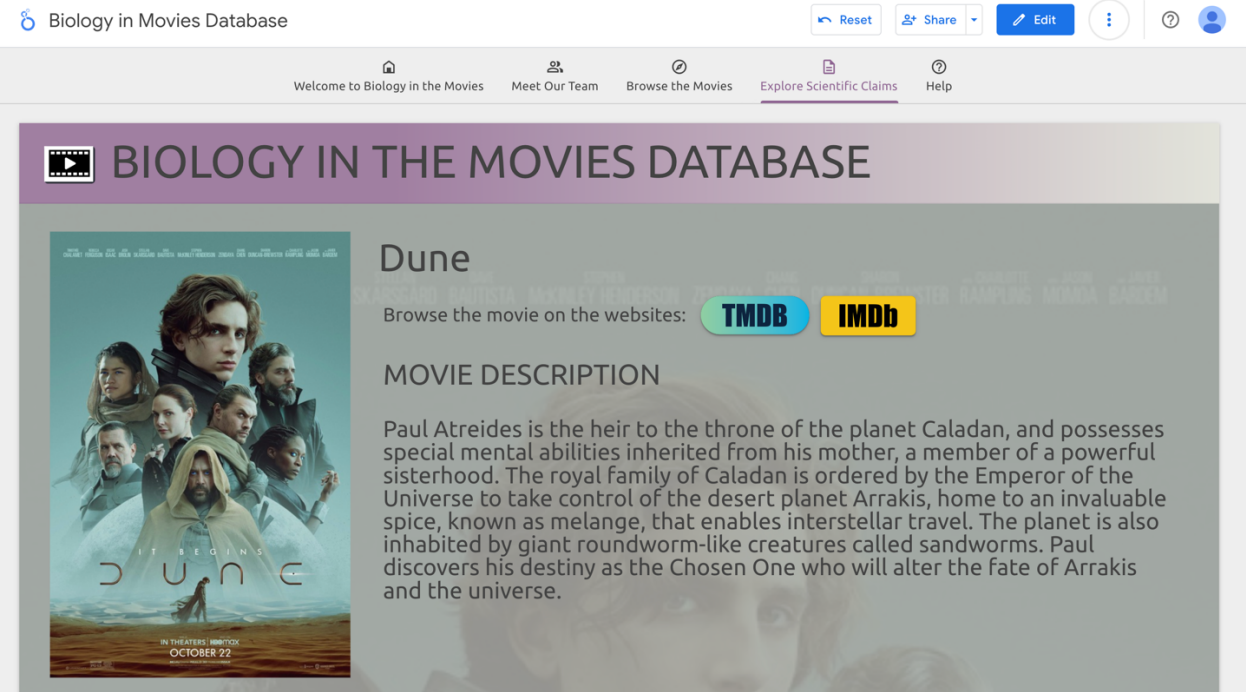


Figure 2. Biology in Movies Report (Looker Studio).

Bibliography

¹*University of Toronto Scarborough Digital Scholarship Unit (DSU)*. Web. 15 Apr. 2024.

< <https://digital.utsc.utoronto.ca> >